

Exam Date & Time: 09-Jan-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - JANUARY 2024

SUBJECT: ECE 3152- LINEAR CONTROL THEORY

LINEAR CONTROL THEORY [ECE 3152]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

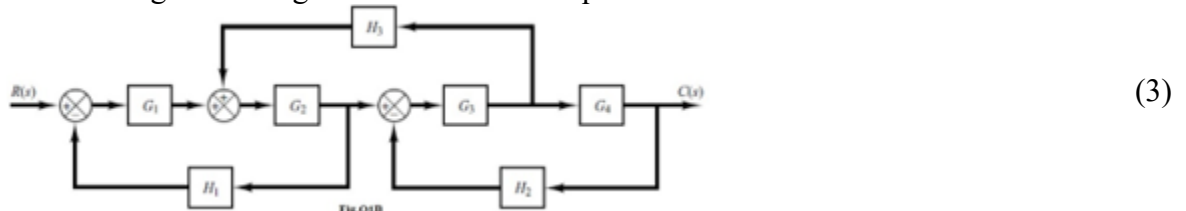
Missing data may be suitably assumed.

- 1A) The open loop transfer function of a unity feedback system is given by

$$G(s) = \frac{K}{s(s^2 + 4s + 11)} \quad (5)$$

Sketch the root locus for the given system and comment on the stability in terms of system gain K.

- 1B) Modify the block diagram shown in Fig.Q1B and develop the closed loop transfer function using block diagram reduction techniques.

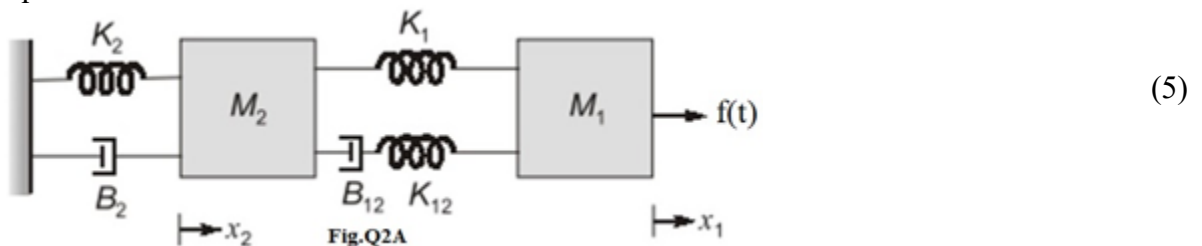


- 1C) A unity feedback system is characterized by an open-loop transfer function,

$$G(s) = \frac{k}{s(s+10)} \quad (2)$$

Determine the gain K so that the system will have damping ratio of 0.5. For this value of K determine the settling time and peak overshoot.

- 2A) Identify the differential equations governing the mechanical behaviour of the system shown in Fig.Q2A. Draw the FI and FV analogous circuits along with mesh and node equations.



- 2B) For the system shown in Fig.Q2B, determine the value of 'K' and 'a' to satisfy the following frequency domain specifications and thereby calculate the settling time and bandwidth of the system for the same values of 'K' and 'a'. (3)

i) $M_r = 1.04$

ii) $\omega_r = 11.55 \text{ rad/sec}$

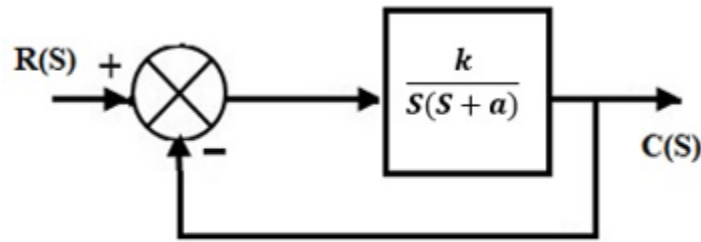


Fig.Q2B

- 2C) A unity feedback system has the forward path transfer function, The input $r(t) = 2 + 6t$ is applied to the system with forward path transfer function

$$G(S) = \frac{K(2s+1)}{s(5s+1)(s+1)^2} \quad (2)$$

Determine the value of K, such that steady state error is equal to 0.1

- 3A) A system has $G(S)H(S) = \frac{K(S+1)}{S(1+TS)(1+2S)}$. The parameters K and T represent x-axis and y-axis respectively. Identify and highlight the region on the graph where the system is stable using Routh Hurwitz criteria. (5)

- 3B) Examine whether the system characterized by the differential equation $\ddot{Y} + 5\dot{Y} + 10Y + 4Y + U = 0$ is controllable and observable using Kalman's method. (3)

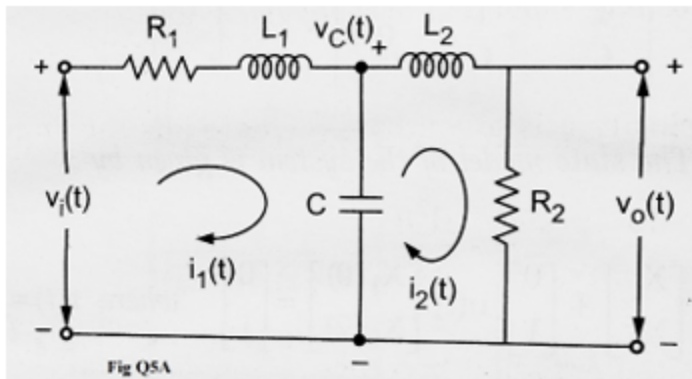
- 3C) For the system represented by the state equation $\dot{X} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} X(t) + \begin{bmatrix} 1 \\ 1 \end{bmatrix} r(t)$ (2)
determine the State Transition Matrix.

- 4A) Sketch the Nyquist plot for a unity feedback system with OLTF $G(S)H(S) = \frac{10(S+3)}{S(S-1)}$ and comment on the stability of the system. (5)

- 4B) Derive an expression for resonant peak M_r of a second order system in terms of its damping ratio when subjected to a pure sinusoidal input (3)

- 4C) Derive an expression for the maximum overshoot of a second order underdamped system when subjected to a unit step input (2)

- 5A) Represent the circuit shown in Fig.Q5A in state space domain using physical variable form. (5)

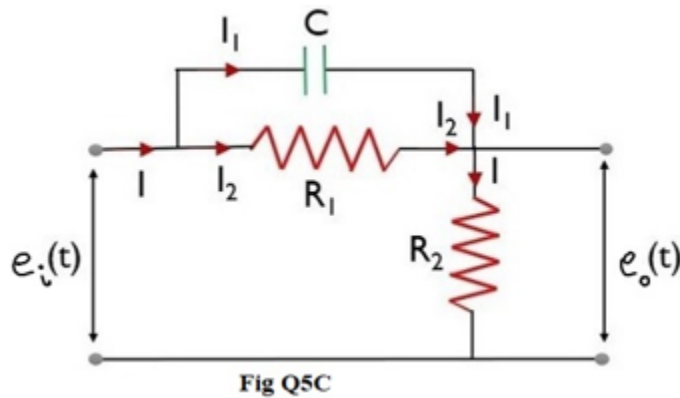


- 5B) A PD controller is cascaded to the forward path of a unity feedback system with plant

transfer function
$$G_P(S) = \frac{16}{S(S+3.5)}$$
 If the proportional gain is 2, estimate the value of K_D for the system to have a damping ratio of 0.9. Also determine the settling time (3)

- 5C) Circuit of a Phase-Lead Compensator is shown in Fig.Q5C where

$R_1=1k\Omega$ $R_2=2k\Omega$ and $C= 200\mu F$. Identify the frequency where the maximum phase lead occurs. Also locate pole and zero of the compensator on the S plane.



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